

Forecasting Ginger Weight Loss During the Drying Process

Team Members:

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Objective:

- To **forecast the percentage of weight (water) lost** by ginger during the drying process using historical production data.
- To **reduce the waiting time for farmer payments** by predicting weight loss earlier in the process, eliminating the need to wait for the full drying and classification stage.

Question:

- What is the percentage of weight that ginger loses during the drying process, given information about its farm of origin and quality classification characteristics?

Supervised or Unsupervised:

This project will use **supervised learning** because we have labeled data in the form of the percentage of lost weight, which is determined after the classification process and it also depends on the farm that the product comes from.

Description of the Data:

✓ Variables:

- **Percentage of lost weight (numerical):** Percentage of weight lost during drying, primarily due to water loss.
- **Farm (categorical):** Farm where the ginger was produced.
- **Percentage of each category (numerical):** During the classification process, ginger is divided into quality categories, and the percentage (by weight) of each category is recorded.
 - CAT
 - Industrial
 - Domestic
 - Discarded
 - Shrinkage

✓ **Sample Size:** Approximately 5000 records from 2024 and 2025.

✓ **Sources:** Internal production and quality control records from the company.

Comments or Concerns:

- **Missing Values:** There may be missing values in some of the variables, particularly if certain samples lack complete quality data. These missing values will need to be handled, either through imputation or exclusion, based on the extent of the missing data.
- **Data Filtering:** Only ginger batches that successfully pass the reception and quality inspection process will be included, as rejected products do not complete the drying process and therefore do not have valid weight-loss measurements.